

ECO102: Introduction to Macroeconomics

Lecture 16: Money, The Price Level, and Inflation Contd.

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The Fed's Policy Tools

- The Fed influences the quantity of money and interest rates by adjusting the quantity of reserves available to the banks and the reserves the banks must hold
- The Fed utilizes three tools to achieve its goals, they are:
- ***Open Market Operations:*** An open market operation is the purchase or sale of securities by the Fed in the loanable funds market

When the Fed buys securities, it pays for them with newly created bank reserves. When the Fed sells securities, the Fed is paid with reserves held by banks.

By changing the quantity of bank reserves, the Fed changes the quantity of monetary base, which influences the quantity of money

The Fed's Policy Tools

- ***Last Resort Loans:*** The Fed is the ***lender of last resort***, which means that if a bank is short of reserves, it can borrow from the Fed. But the Fed sets the interest rate on last resort loans and this interest rate is called the discount rate
- ***Required Reserve Ratio:*** The required reserve ratio is the minimum percentage of deposits that depository institutions are required to hold as reserves.

If the Fed requires the banks to hold more reserves, the banks must cut their lending

How Banks Create Money

- Remember, money is both currency and bank deposits. What banks create is deposits, and they do so by making loans
- Also remember that, bank deposits and currencies are both money
- Three factors limit the quantity of loans and deposits that the banking system can create through transactions:
 - *The Monetary Base*
 - *Desired Reserves*
 - *Desired Currency Holding*

How Banks Create Money

- ***The Monetary Base:*** The size of the monetary base limits the total quantity of money that the banking system can create
- ***Desired Reserves:*** A bank's desired reserves are the reserves that it plans to hold. They contrast with a bank's required reserves, which is the minimum quantity of reserves that a bank must hold

The quantity of desired reserves depends on the level of deposits and is determined by the ***desired reserve ratio***, the ratio of reserves to deposits that the banks plan to hold

- ***Desired Currency Holding:*** The proportions of money held as currency and bank deposits depend on how households and firms choose to make payments

When banks make loans that increase deposits, some currency leaves the banks, meaning the banking system leaks reserves. We call the leakage of bank reserves into currency the currency drain, and we call the ratio of currency to deposits the ***currency drain ratio***

The Money Creation Process

- The money creation process begins with an increase in the monetary base, which occurs if the Fed conducts an open market operation in which it buys securities from banks and other institutions. Leading to the bank having excess reserves. A bank's **excess reserves** are its actual reserves minus its desired reserves
- There is sequential process of money creation consists of eight steps:
 1. Banks have excess reserves
 2. Banks lend excess reserves
 3. The quantity of money increases
 4. New money is used to make payments
 5. Some of the new money remains on deposit.
 6. Some of the new money is a currency drain
 7. Desired reserves increase because deposits have increased
 8. Excess reserves decrease

The Money Market

- There is no limit to the amount of money we would like to receive in payment for our labour or as interest on our savings
- But there is a limit to how big an inventory of money we would like to hold and neither spend nor use to buy assets that generate an income
- The quantity of money demanded is the inventory of money that people plan to hold on any given day. It is the quantity of money in our wallets and in our deposit accounts at banks
- The quantity of money held must equal the quantity supplied, and the forces that bring about this equality in the money market have powerful effects on the economy

Influences on Money Holding

- The quantity of money that people plan to hold depends on four main factors:
- ***The price level:*** The quantity of money measured in dollars is ***nominal money***. The quantity of nominal money demanded is proportional to the price level

If the price level rises by 10 percent, people hold 10 percent more nominal money than before, other things remaining the same

The quantity of money measured in constant dollars (for example, in 2005 dollars) is real money. Real money is equal to nominal money divided by the price level and is the quantity of money measured in terms of what it will buy. The quantity of real money demanded is independent of the price level

Influences on Money Holding

- ***The nominal interest rate:*** The higher the opportunity cost of holding money, other things remaining the same, the smaller is the quantity of real money demanded.

The nominal interest rate on other assets minus the nominal interest rate on money is the opportunity cost of holding money

The interest rate that you earn on currency and checking deposits is zero. So the opportunity cost of holding these items is the nominal interest rate on other assets such as a savings bond or Treasury bill

Influences on Money Holding

- **Real GDP:** The quantity of money that households and firms plan to hold depends on the amount they are spending. The quantity of money demanded in the economy as a whole depends on aggregate expenditure, i.e. real GDP
- **Financial innovation:** Technological change and the arrival of new financial products influence the quantity of money held. Financial innovations include:
 1. Daily interest checking deposits
 2. Automatic transfers between checking and saving deposits
 3. Automatic teller machines
 4. Credit cards and debit cards
 5. Internet banking and bill paying

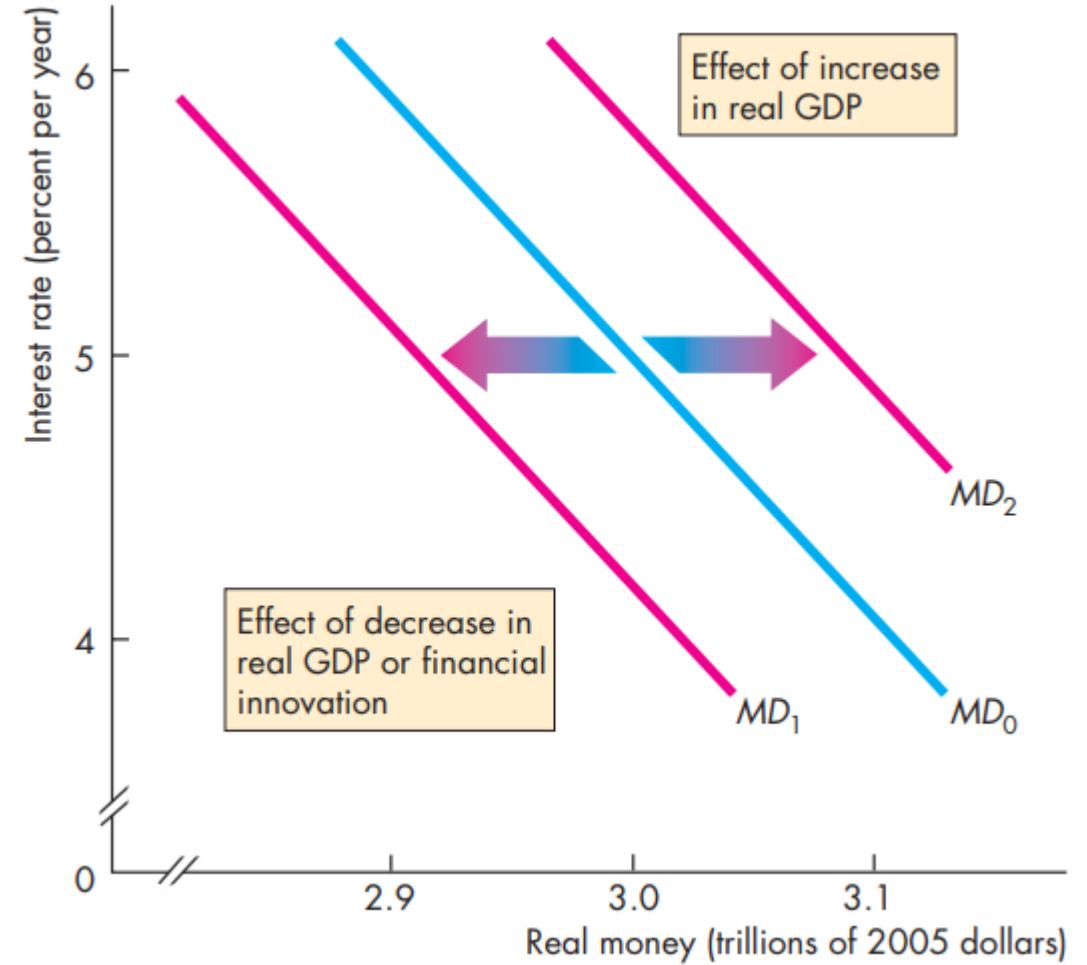
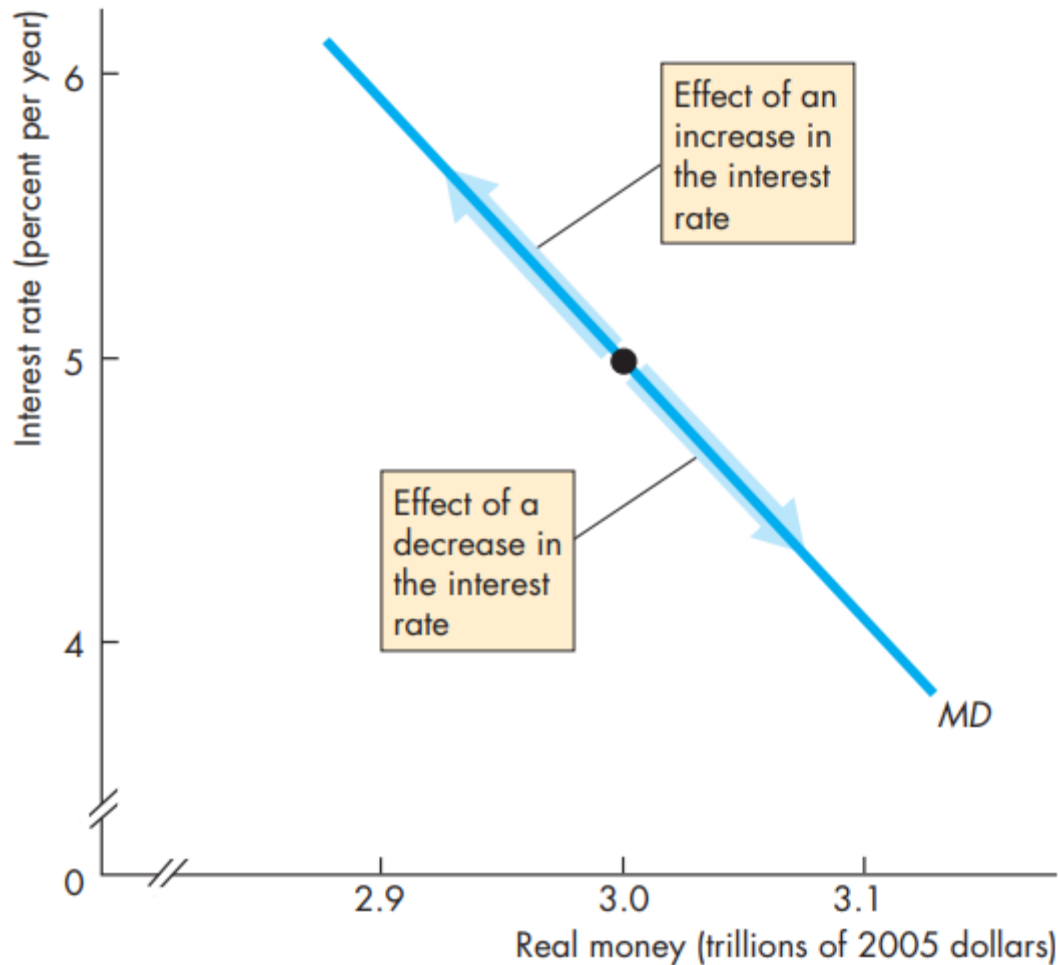
The Demand for Money

- The demand for money is the relationship between the quantity of real money demanded and the nominal interest rate when all other influences on the amount of money that people wish to hold remain the same

Shifts in the Demand for Money Curve

- A change in real GDP or financial innovation changes the demand for money and shifts the demand for money curve
- A decrease in real GDP decreases the demand for money and shifts the demand for money curve leftward; increase in real GDP has the opposite effect
- The influence of financial innovation on the demand for money curve is more complicated. But generally, financial innovation decreases the demand for money

The Demand for Money



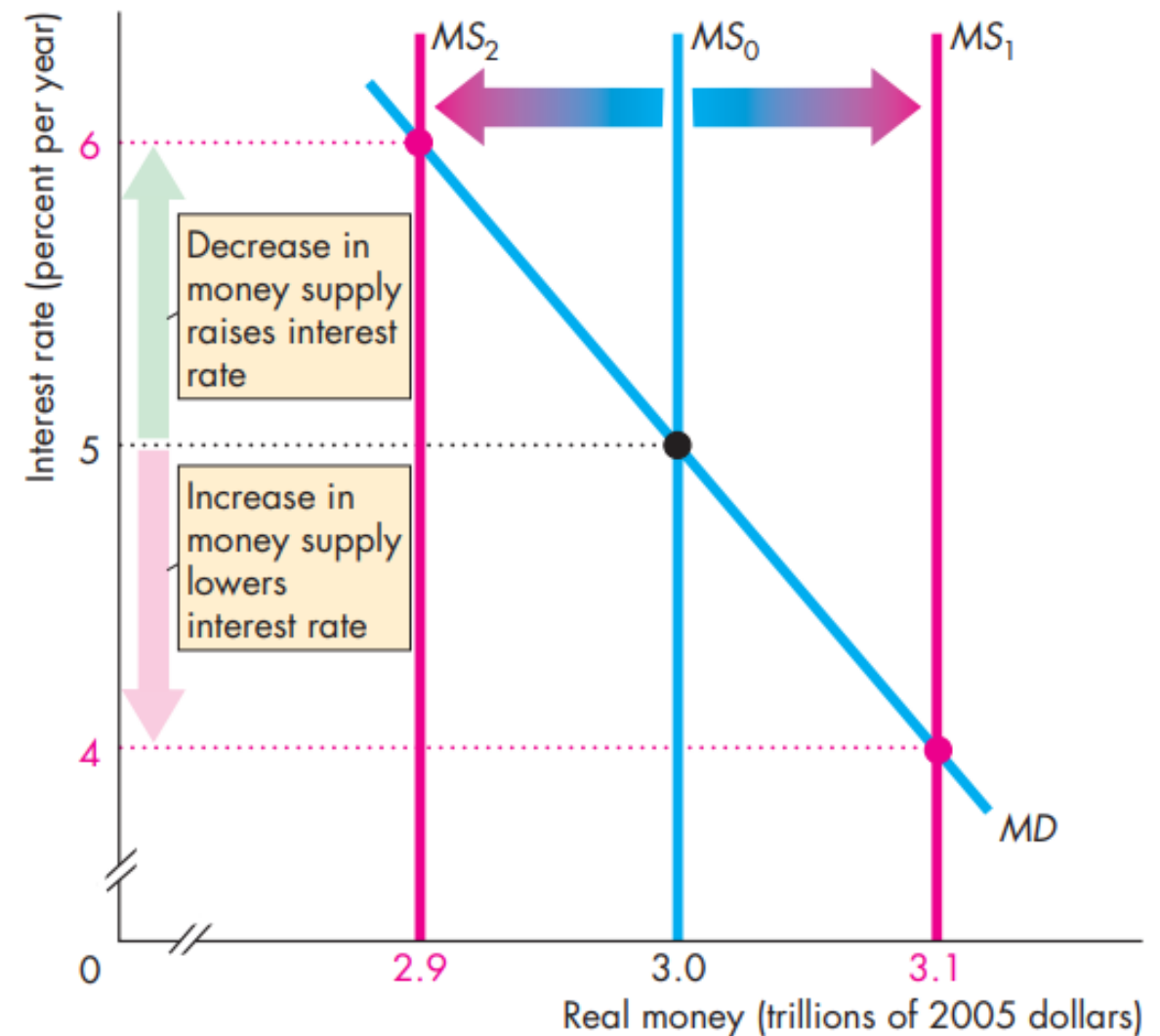
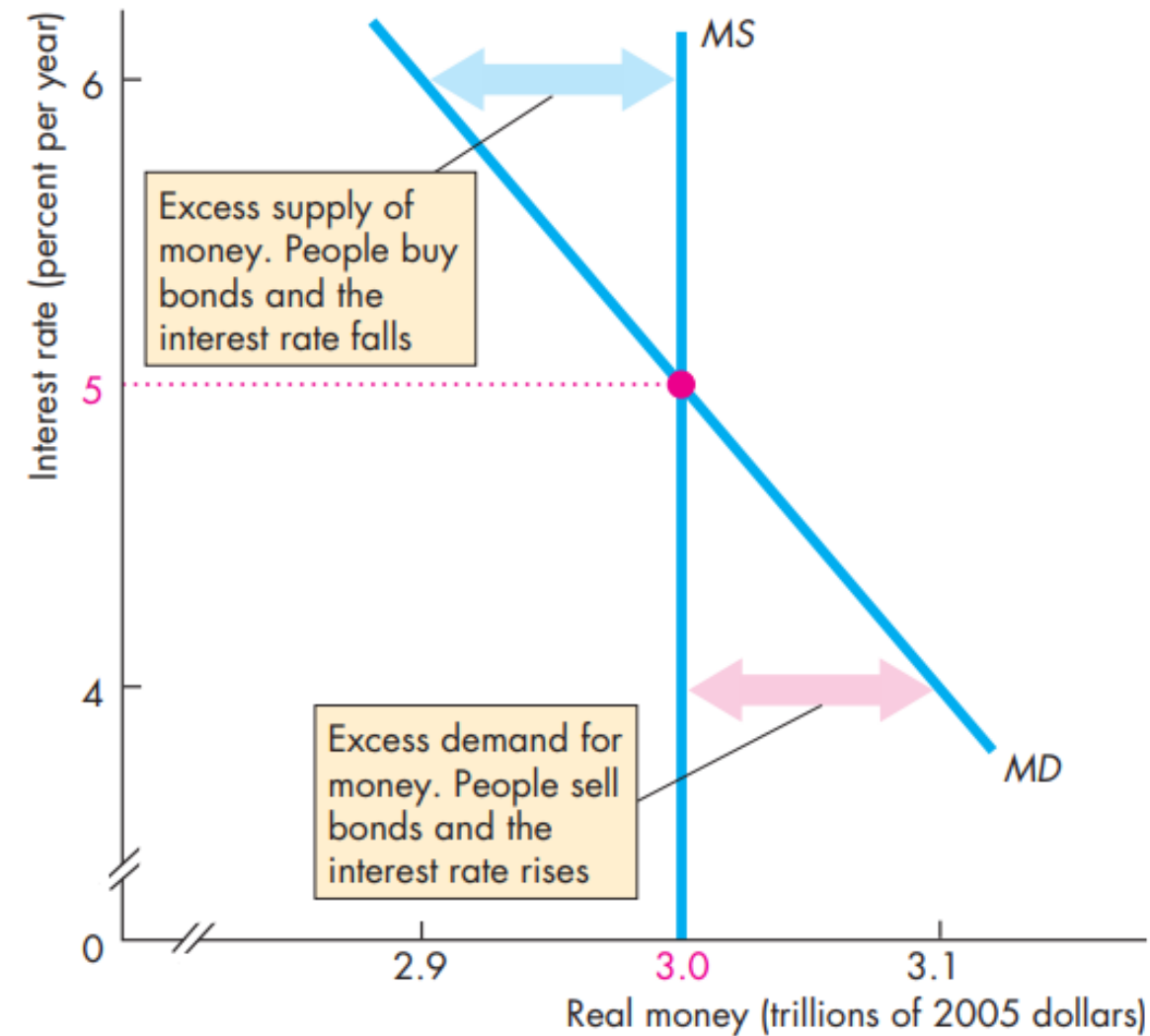
Money Market Equilibrium

- Money market equilibrium occurs when the quantity of money demanded equals the quantity of money supplied. The adjustments that occur to bring money market equilibrium are fundamentally different in the short run and the long run
- ***Short-run Equilibrium:*** Starting from a short-run equilibrium, if the Fed increases the quantity of money, people find themselves holding more money than the quantity demanded. With a surplus of money holding, people enter the loanable funds market and buy bonds. The increase in demand for bonds raises the price of a bond and lowers the interest rate.
- The opposite happens, when the Fed decreases the quantity of money

Money Market Equilibrium

- ***Long-run Equilibrium:*** When the inflation rate equals the expected (or forecasted) inflation rate and when real GDP equals potential GDP, the money market, the loanable funds market, the goods market, and the labor market are in long-run equilibrium, meaning the economy is in long-run equilibrium
- If in long-run equilibrium, the Fed increases the quantity of money, eventually a new long-run equilibrium is reached in which nothing real has changed. Real GDP, employment, the real quantity of money, and the real interest rate all return to their original levels. This happens because the price level rises by the same percentage as the rise in the quantity of money.

Money Market Equilibrium



The Quantity Theory of Money

- The quantity theory of money explains the long-run adjustment of the price level
- This theory proposes that in the long run, an increase in the quantity of money brings an equal percentage increase in the price level
- The **velocity of circulation** is the average number of times a dollar of money is used annually to buy the goods and services that make up GDP
- The **velocity of circulation**, V , is determined by the equation:

$$V = \frac{PY}{M}$$

Where, M = Quantity of Money; P = Price Level; & Y = Real GDP

- The equation of exchange tells us:

$$MV = PY$$

- We also get:

$$P = M \left(\frac{V}{Y} \right)$$

The Quantity Theory of Money

- (V/Y) is independent of M ; meaning a change in M brings a proportional change in P
- We can also express the equation of exchange in growth rates:

$$\text{Money Growth Rate} + \text{Rate of Velocity Change} = \text{Inflation Rate} + \text{Real GDP Growth Rate}$$

- Solving this we get:

$$\text{Inflation Rate} = \text{Money Growth Rate} + \text{Rate of Velocity Change} - \text{Real GDP Growth Rate}$$

- In the long run, the rate of velocity change is approximately zero, meaning:

$$\text{Inflation Rate} = \text{Money Growth Rate} - \text{Real GDP Growth Rate}$$